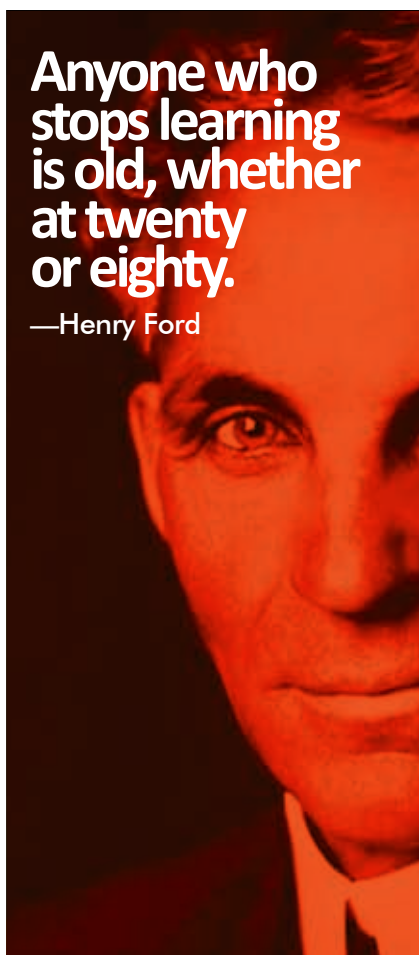




Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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This is where SIM-connected IoT plays a foundational role. eSIM and iSIM solutions, particularly those compliant with SGP.32 and SGP.42 standards, offer secure, scalable, and remotely manageable connectivity for IoT devices. These technologies support in-factory profile provisioning and advanced remote SIM provisioning, ensuring devices are secure and authenticated from the moment they are manufactured, while reducing the number of stock keeping units, thereby providing manufacturers and their customers with greater global flexibility and scalability.

Built-in security from the start

With eSIMs and iSIMs, connectivity and security are not afterthoughts; they are built in from the beginning. These SIMs are tamper-resistant by design and equipped with secure elements that protect against both physical and remote interference. They enable encrypted data transmission and support digital signatures that verify data authenticity at every step.

Through features such as ECC, devices can encrypt and decrypt data using mathematically complex key pairs, making it extremely difficult for attackers to access or alter information. This level of security is especially important in AI-powered environments, where even a small manipulation of training data can lead to inaccurate predictions and flawed system behaviour.

Strengthening data integrity for reliable AI

AI systems are only as reliable as the data they are trained on and operate on. In the world of distributed IoT networks, where countless sensors and devices continuously generate and transmit data, ensuring data integrity is non-negotiable.

Blockchain plays a complementary role in this context, providing a decentralised and immutable

ledger that records transactions and data exchanges across devices. This enhances trust and ensures that once data is validated, it cannot be altered. When combined with SIM-based identity and encryption, organisations can achieve data integrity that is foundational to deploying effective, secure AI.

The identity of things: A trusted ecosystem

SIM connectivity goes beyond securing data; it helps establish a new paradigm: the identity of things. Every device, sensor, and endpoint in an IoT ecosystem can be uniquely identified, authenticated, and managed through its SIM profile. This identity-centric approach ensures not only secure connectivity but also controlled access, scalability, and compliance with evolving industry standards.

The SGP.32 and SGP.42 specifications further reinforce this by providing a unified framework for secure remote provisioning and SIM lifecycle management across diverse IoT use cases.

Laying the groundwork for an intelligent, trusted future

Together, AI and IoT are enabling a future of intelligent automation, from predictive maintenance and supply chain optimisation to smart cities and precision agriculture. However, this future depends on the trustworthiness of the underlying data.

Organisations must prioritise secure-by-design architectures with SIM-enabled connectivity to ensure that AI-powered decisions are accurate, timely, and safe. It is not only about smarter systems, but also about smarter, safer, and more secure systems.

When AI meets IoT, trust is non-negotiable. With SIM technologies at the core, the industry is building a future where connected intelligence can thrive securely and sustainably. **EFY**